Seth Ricks | ECEN 470 | 10/13/2026 | PS04: 4.2(a)(b)(c), 4.15(a)(b), 4.44(a)(b)(c)(d)(e)(f)(g)

(4.2) We define the sensitivity of a transfer function G to one of its parameters K as the ratio of percent change in G to percent change in K

$$S_K^G = \frac{dG/G}{dK/K} = \frac{d\ln G}{d\ln K} = \frac{K}{G}\frac{dG}{dK}$$

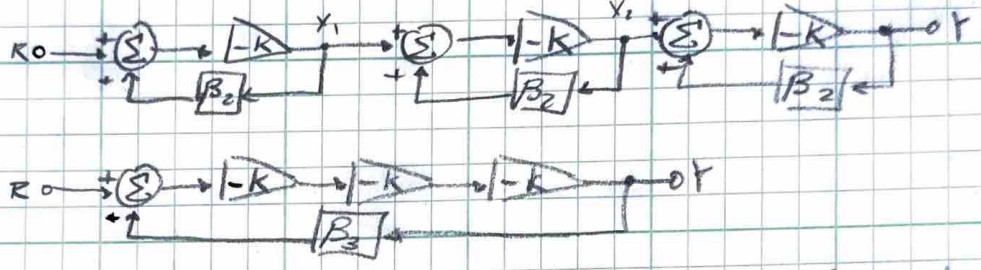
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Compare topologies shown in figures for connecting three amplifier stages with a gain of $-K$ into a single amplifier with gain of -10.

1.)



2.)



a.) For each topology, compute β_i so if $K=10$, $Y=-10R$

b.) For each topology, compute S_K^G when $G_1 = \frac{Y}{R}$. (Use the respective β_i values found in part (a).) Which case is least sensitive?

c.) Compute sensitivities of systems to β_2 and β_3. Using results, comment on relative need for precision in sensors and actuators

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a.) $Y = +R\beta_1 K^3 = +10R$

$\beta_1 (1000) = 10$

$\boxed{\beta_1 = \frac{1}{100}}$

$$\frac{X_1}{R} = \frac{-K}{1+(+K)(+\beta_2)}$$

$$X_1 = \frac{-K}{1+K\beta_2} R$$

$$\frac{X_2}{X_1} = \frac{-K}{1+K\beta_2}$$

$$X_2 = \frac{-K}{1+K\beta_2} X_1$$

$$\frac{Y}{X_2} = \frac{-K}{1+K\beta_2}$$

$$Y = \frac{-K}{1+K\beta_2} X_2$$

$$Y = \frac{-K}{1+K\beta_2}\left[\frac{-K}{1+K\beta_2}\left[\frac{-K}{1+K\beta_2} R\right]\right]$$

$$Y = +\left(\frac{K}{1+K\beta_2}\right)^3 R = +10R$$

$$\frac{-K}{1+K\beta_2} = \sqrt[3]{10}$$

$$-K = \sqrt[3]{10} + \sqrt[3]{10}\,K\beta_2$$

$$\beta_2 = \frac{K - \sqrt[3]{10}}{(\sqrt[3]{10})(10)} = 0.36$$

$\boxed{\beta_2 = 0.36}$

$$\frac{Y}{R} = \frac{-K^3}{1+(+K^3)(+\beta_3)}$$

$$Y = \frac{+K^3}{1+K^3\beta_3} R = +10R$$

$$K^3 = 10 + 10K^3\beta_3$$

$$\frac{K^3 - 10}{10K^3} = \beta_3$$

$\boxed{\beta_3 = 0.099}$

b.) $S_K^G = ?$ when $G = \frac{Y}{R}$

$$S_K^G = \frac{K}{G}\frac{\partial G}{\partial K}$$

1.) $Y = -R\beta_1 k^3$

$G = \frac{-R\beta_1 k^3}{R} = -\beta_1 k^3$

$$S_K^G = \frac{K}{-\beta_1 k^3}(-3\beta_1 k^2) = \frac{+3\beta_1 k^3}{+\beta_1 k^3} = \boxed{3}$$

2.) $Y = -\left(\frac{K}{1+k\beta_2}\right)^3 R$

$$G = \frac{-\left(\frac{K}{1+k\beta_2}\right)^3 R}{R} = -\left(\frac{K}{1+k\beta_2}\right)^3 = -\left[\frac{k^3}{(1+k\beta_2)^3}\right]$$

$$S_K^G = \frac{K}{-\left(\frac{K}{1+k\beta_2}\right)^3}\left[-\left[\frac{(1+k\beta_2)^3(3k^2) - k^3(3\beta_2(1+k\beta_2)^2)}{(1+k\beta_2)^6}\right]\right]$$

$$= \frac{3}{k\beta_2 + 1} = \boxed{0.646}$$

3.) $Y = \frac{-k^3 R}{1+\beta_3 k^3}$

$$G = \frac{Y}{R} = \frac{-k^3}{1+\beta_3 k^3}$$

$$S_K^G = \frac{-K(1+\beta_3 k^3)}{k^3}\left[\frac{(1+\beta_3 k^3)(3k^2) - (k^3)(3\beta_3 k^2)}{(1+\beta_3 k^3)^2}\right]$$

$$= \frac{3}{\beta_3 k^3 + 1} = \boxed{0.03}$$

1.) $\boxed{3}$

2.) $\boxed{0.646}$

3.) $\boxed{0.03}$ ← #3 is least sensitive

c.) $$S_{\beta_2}^G = \frac{\beta_1}{G}\frac{\partial G}{\partial \beta_2} = \frac{\beta_2}{Y/R}\frac{\partial (Y/R)}{\partial \beta_2} \qquad Y/R = \left(\frac{K}{1+K\beta_2}\right)^3$$

$$= \frac{\beta_2(1+k\beta_2)^3}{k^3} * \frac{-3k}{(1+k\beta_2)^4} = \frac{-3\beta_2 K}{1+k\beta_2} = \frac{-3(0.36)(10)}{1+10(0.36)}$$

$$= \boxed{-2.35}$$

$$S_{\beta_3}^G = \frac{\beta_2}{G}\frac{\partial G}{\partial \beta_3} = \frac{\beta_3}{Y/R}\frac{\partial (Y/R)}{\partial \beta_3} \qquad Y/R = \frac{+k^3}{1+k^3\beta_3}$$

$$= -\frac{\beta_3(1+k^3\beta_3)}{k^3}\frac{k^3}{(1+k^3\beta_3)^2} = \frac{-k^3\beta_3}{(1+k^3\beta_3)} = \boxed{-0.99}$$

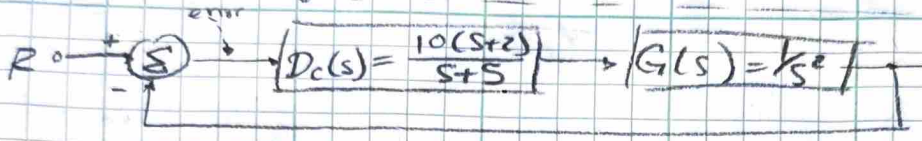
The systems are much more sensitive to the neural than the amplifiers.

(4.15) A controller for a satellite attitude control with TF $G = 1/s^2$ has been designed with a unity feedback structure and has the TF $D_c(s) = \frac{10(s+2)}{s+5}$. | 3/4 |

a.) Find system type for reference tracking and the corresponding error constant for this system.

b.) If a disturbance torque adds to the control so that the input to the process is $u+w$, what is the system type and corresponding error constant with respect to disturbance rejection?

a.)



$e = r - y$; $Y = ED_cG$

$E = R - ED_cG$

$E = \frac{R}{1+D_cG}$

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} \frac{sR}{1+D_cG} = \frac{\lim_{s\to 0} s}{1+\frac{10(s+2)}{(s+5)s^2}} R$$

$$\lim_{s\to 0} \frac{s}{\frac{(s+5)s^2}{(s+5)s^2} + \frac{10(s+2)}{(s+5)s^2}} R = \frac{\lim_{s\to 0} s}{\frac{(s+5)s^2+10(s+2)}{(s+5)s^2}} R$$

input: $\frac{1}{s^{k+1}}$

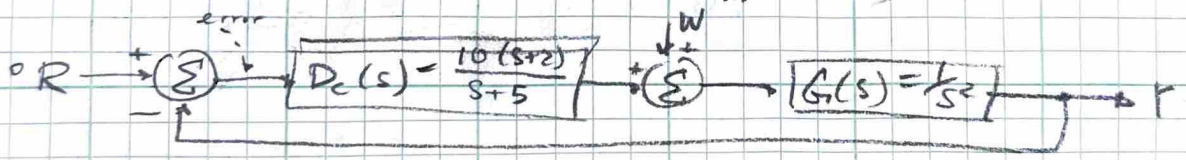
$$\lim_{s\to 0} \frac{s^3(s+5)}{(s+5)s^2+10(s+2)} R$$

$k=0$: $R = 1/s \rightarrow e_{ss} = 0$

$k=1$: $R = 1/s^2 \rightarrow e_{ss} = 0$

$k=2$: $R = 1/s^3 \rightarrow e_{ss} = 5/20 = 1/4$ | type 2 | $K_a = 4$ |

b.)



$Y = (ED_c + W)(G) = EDcG + WG$

$E = R - Y$

$= R - EDcG - WG$

$E + EDcG = R - WG$

$E = \frac{R-WG}{1+DcG} = \frac{R}{1+DcG} - \frac{WG}{1+DcG}$

$$e_{ss} = \lim_{s\to 0} \frac{-sWG}{1+DcG} = \lim_{s\to 0} \frac{-s(1/s^2)}{1+\frac{10(s+2)}{s+5}(1/s^2)} W$$

$$\frac{-\frac{1}{s}}{\frac{s^2(s+5)}{s^2(s+5)} + \frac{10(s+2)}{s^2(s+5)}} = \frac{-1/s}{\frac{s^2(s+5)+10(s+2)}{s^2(s+5)}}$$

$$= -\frac{1}{s} * \frac{s^2(s+5)}{s^2(s+5)+10(s+2)}$$

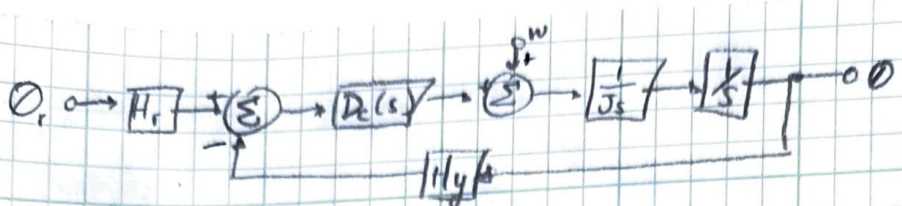
$$= \lim_{s\to 0} \frac{-s(s+5)}{s^2(s+5)+10(s+2)} W$$

$k=0$: $W = 1/s \rightarrow e_{ss} = -5/20 = -1/4$

type 0

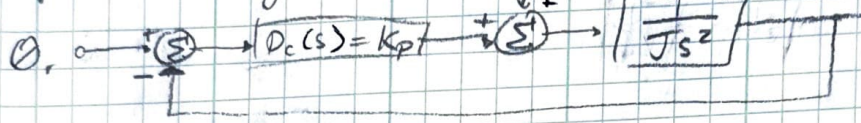
$K_a = -4$

(4.44)



$J = 10$ (spacecraft inertia), $N \cdot m \cdot sec^2/rad$
θ_r = reference satellite attitude, rad
θ = actual satellite attitude, rad
$H_y = 1$ (sensor scale factor), V/rad
$H_r = 1$ (reference sensor scale factor), V/rad
w = disturbance torque, $N \cdot m$

a.) Use proportional control P, $D_c(s) = K_p$, and give range of values for K_p where system will be stable



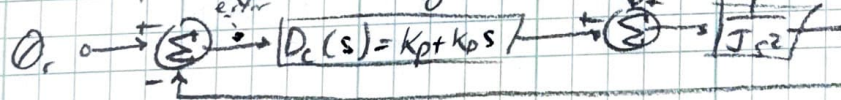
$$\frac{\theta}{\theta_r} = \frac{K_p \frac{1}{Js^2}}{1 + K_p \frac{1}{Js^2}} = \frac{K_p/Js^2}{\frac{Js^2 + K_p}{Js^2}} = \frac{K_p}{Js^2} * \frac{Js^2}{Js^2 + K_p} = \frac{K_p}{Js^2 + K_p} = \frac{K_p}{10s^2 + K_p} = \frac{\frac{K_p}{10}}{s^2 + \frac{K_p}{10}}$$

$$= \frac{K_p}{(s - j\sqrt{\frac{K_p}{10}})(s + j\sqrt{\frac{K_p}{10}})}$$

if $K_p < 0$, $s - j\sqrt{\frac{-K}{10}} = s - j(j\sqrt{\frac{K}{10}}) = s + \sqrt{\frac{K}{10}}$ = unstable
$K_p = 0$, s^2 = unstable
$K_p > 0$, $s - j\sqrt{\frac{K}{10}}$ = imaginary → stable
$s + j\sqrt{\frac{K}{10}}$ = "

$\boxed{K_p > 0}$

b.) Use PD control, $D_c(s) = K_p + K_D s$, find system type and error constant w/ respect to reference inputs



$E = R - Y$
$Y = E D_c G$
$E = R - E D_c G$

$$E = \frac{R}{1 + D_c G} = \frac{R}{1 + \frac{K_p + K_p s}{Js^2}} = \frac{R}{\frac{Js^2 + K_p + K_p s}{Js^2}} = \frac{Js^2 R}{Js^2 + K_p + K_p s}$$

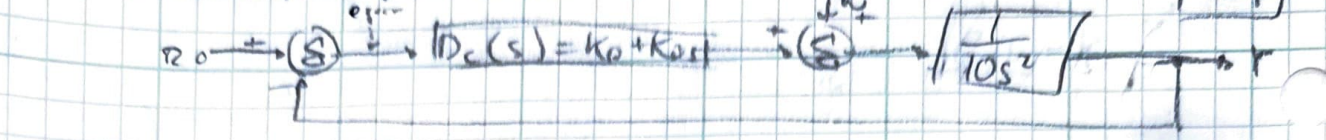
$$e_{ss} = \lim_{s \to 0} \frac{10s^3}{10s^2 + K_p + K_p s} R$$

$k = 0$: $R = \frac{1}{s}$ → $e_{ss} = 0$
$k = 1$: $R = \frac{1}{s^2}$ → $e_{ss} = 0$
$k = 2$: $R = \frac{1}{s^3}$ → $e_{ss} = 10/K_p$

$\boxed{e_{ss} = \frac{10}{K_p}, \text{ type 2}}$

c.) PD control, let $D_c(s) = k_p + k_D s$, find system type & error constant w/ respect to disturbance input

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$E = R - Y$

$Y = (E D_c + W) G$

$E = \frac{R}{1+D_cG} - \frac{WG}{1+D_cG}$

$$\frac{-WG}{1+D_cG} = \frac{-W/10s^2}{1+\frac{k_p+k_Ds}{10s^2}} = \frac{-W/10s^2}{\frac{10s^2+k_p+k_Ds}{10s^2}}$$

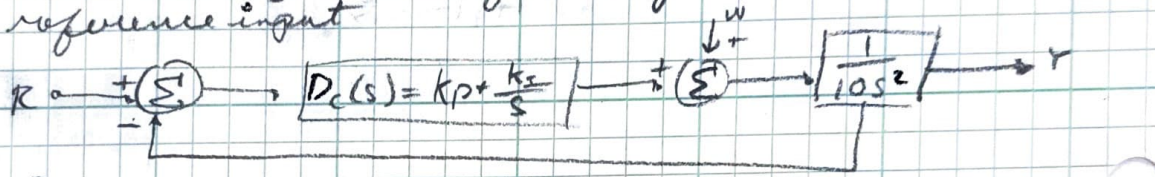
$$= -\frac{W}{10s^2} \cdot \frac{10s^2}{10s^2+k_p+k_Ds} = \frac{-1}{10s^2+k_p+k_Ds} W$$

$$e_{ss} = \lim_{s\to 0} \frac{-s}{10s^2+k_p+k_Ds} W$$

$k=0$: $W = \frac{1}{s} \rightarrow e_{ss} = \frac{-1}{k_p}$

$\boxed{e_{ss} = \frac{-1}{k_p}, \text{ type } 0}$

d.) Use PI, let $D_c(s) = k_p + \frac{k_I}{s}$, find system type & error constant w/ respect to reference input



$$E = \frac{R}{1+D_cG} = \frac{R}{1+\frac{k_p+\frac{k_I}{s}}{10s^2}} = \frac{R}{1+\frac{k_ps+k_I}{10s^3}} = \frac{R}{\frac{10s^3+k_ps+k_I}{10s^3}}$$

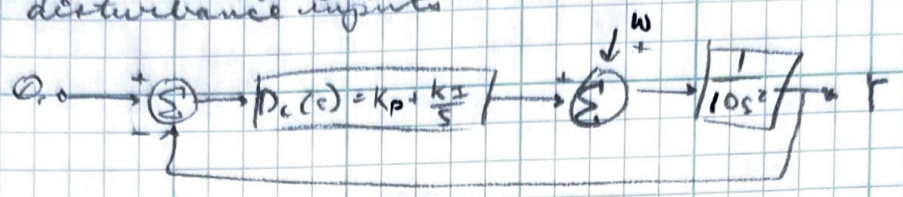
$$= \frac{10s^3}{10s^3+k_ps+k_I} R$$

$$e_{ss} = \lim_{s\to 0} \frac{10s^4}{10s^3+k_ps+k_I} R$$

$k=3$: $R = \frac{1}{s^4} \rightarrow \boxed{e_{ss} = \frac{10}{k_I}, \text{ type } 3}$

e.) PI, $D_c(s) = K_p + \frac{K_I}{s}$, find system type & error constant w/ respect to disturbance inputs

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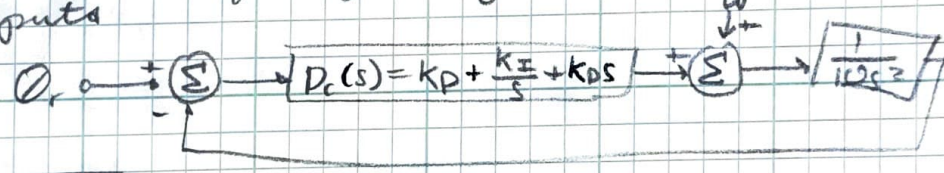
$$E = \frac{-WG}{1+D_cG} = \frac{-W/10s^2}{1+\frac{K_p+\frac{K_I}{s}}{10s^2}} = \frac{-W/10s^2}{1+\frac{K_ps+K_I}{s}} = \frac{-W/10s^2}{\frac{10s^3}{10s^3}+\frac{K_ps+K_I}{10s^3}}$$

$$= \frac{-W}{10s^2} \cdot \frac{10s^3}{10s^3+K_ps+K_I} = \frac{-W}{10s^3+K_ps+K_I}$$

$$e_{ss} = \frac{-s^2}{10s^3+K_ps+K_I} W$$

K: 1 $W = \frac{1}{s^2} \rightarrow$ $e_{ss} = \frac{-1}{K_I}$, type 1

f.) PID, $D_c(s) = K_p + \frac{K_I}{s} + K_Ds$, find system type & error constant to reference inputs



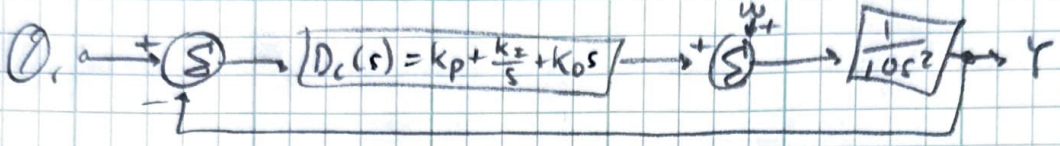
$$E = \frac{R}{1+D_cG} = \frac{R}{1+\frac{(K_p+\frac{K_I}{s}+K_Ds)}{10s^2}}$$

$$= \frac{R}{1+\frac{K_ps+K_I+K_Ds^2}{s}} = \frac{R}{\frac{10s^3}{10s^3}+\frac{K_ps+K_I+K_Ds^2}{10s^3}} = \frac{10s^3}{10s^3+K_ps+K_I+K_ps^2} R$$

$$e_{ss} = \lim_{s\to 0} = \frac{10s^4}{10s^3+K_ps+K_I+K_ps^2}$$

K: 3 $W = \frac{1}{s^4} \rightarrow$ $e_{ss} = \frac{10}{K_I}$, type 3

9.) PID, $D_c(s) = (K_p + \frac{K_I}{s} + K_D s)$, determine type & error constants w/ respect to disturbance input. 7/7



$$E = \frac{-WG}{1+D_cG} = \frac{-W/10s^2}{1+\frac{K_p+\frac{K_I}{s}+K_D s}{10s^2}} = \frac{-W/10s^2}{\frac{10s^2+K_p s+K_I+K_D s^2}{10s^3}}$$

$$= \frac{-W}{10s^2} \cdot \frac{10s^2}{10s^2+K_p s+K_I+K_p s^2}$$

$$= \frac{-W}{10s^2+K_p s+K_I+K_p s^2}$$

$$e_{ss} = \lim_{s\to 0} \frac{-10s^2}{10s^2+K_p s+K_I+K_p s^2} W$$

$K=1$: $W = \frac{1}{s^2} \rightarrow$ $e_{ss} = \frac{-1}{K_I}$, type 1

Is e_{ss} something at error constant?

↳ same or different for disturbance/reference